

Installing Linux Headers on DEBIX R3576-01

NOTE:

The factory Linux system on the DEBIX R3576-01 does not include Linux headers by default. If you need to compile kernel modules, you must manually install the corresponding header package. This document provides the installation steps and a verification method.

Version Information: Linux version 6.1.84

1. Preparation

- Hardware: DEBIX R3576-01 development board with Linux already booted.
- Software package: [linux-headers-6.1-arm64_arm64.deb \(for kernel 6.1.84\)](#).
- File transfer tools: USB flash drive, SCP, TFTP, or similar methods to copy the file to the board.
- (Optional) Flashing tool: You may use RKDevTool instead of FactoryTool for image flashing.

2. Install Linux Headers

- (1) Copy the file `linux-headers-6.1-arm64_arm64.deb` to any directory on the development board (for example `/home/debian`).
- (2) Log in to the board terminal, switch to the directory where the file is located, and run the following command to install the Linux headers:

```
dpkg -i linux-headers-6.1-arm64_arm64.deb
```

3. Verify the Installation

You can verify whether the headers are installed correctly by compiling a simple kernel module.

- (1) Create a working directory

On the development board, create a working directory (for example `~/test_module`) and place the following two files inside it: **Makefile** and **test_hello.c**

Makefile

```
obj-m := test_hello.o

KDIR := /lib/modules/$(shell uname -r)/build
PWD := $(shell pwd)

default:
    $(MAKE) -C $(KDIR) M=$(PWD) modules

clean:
    $(MAKE) -C $(KDIR) M=$(PWD) clean
```

test_hello.c

```
#include <linux/init.h>
#include <linux/module.h>

MODULE_LICENSE("GPL");

static int __init hello_init(void) {
    printk(KERN_INFO "Hello, kernel headers test!\n");
    return 0;
}

static void __exit hello_exit(void) {
    printk(KERN_INFO "Goodbye, kernel headers test!\n");
}

module_init(hello_init);
module_exit(hello_exit);
```

(2) Compile the module

Run the following command in the `~/test_module` directory:

```
make
```

If the compilation is successful, the file `test_hello.ko` will be generated.

(3) Load and test the module

```
sudo insmod test_hello.ko      # Load module  
sudo rmmod test_hello          # Remove module
```

Check the kernel log using:

```
dmesg | tail
```

You should see the following messages:

```
Hello, kernel headers test!  
Goodbye, kernel headers test!
```

If these messages appear, it indicates that the Linux headers have been installed correctly and kernel module compilation works properly.